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Using In Vivo Cine and 3D Multi-Contrast MRI to Determine Human Atherosclerotic Carotid Artery Material Properties and Circumferential Shrinkage Rate and Their Impact on Stress/Strain Predictions

In vivo magnetic resonance image (MRI)-based computational models have been introduced to calculate atherosclerotic plaque stress and strain conditions for possible rupture predictions. However, patient-specific vessel material properties are lacking in those models, which affects the accuracy of their stress/strain predictions. A noninvasive approach of combining in vivo Cine MRI, multicontrast 3D MRI, and computational modeling was introduced to quantify patient-specific carotid artery material properties and the circumferential shrinkage rate between vessel in vivo and zero-pressure geometries. In vivo Cine and 3D multicontrast MRI carotid plaque data were acquired from 12 patients after informed consent. For each patient, one nearly-circular slice and an iterative procedure were used to quantify parameter values in the modified Mooney-Rivlin model for the vessel and the vessel circumferential shrinkage rate. A sample artery slice with and without a lipid core and three material parameter sets representing stiff, median, and soft materials from our patient data were used to demonstrate the effect of material stiffness and circumferential shrinkage process on stress/strain predictions. Parameter values of the Mooney-Rivlin models for the 12 patients were quantified. The effective Young's modulus (YM, unit: kPa) values varied from 137 (soft), 431 (median), to 1435 (stiff), and corresponding circumferential shrinkages were 32%, 12.6%, and 6%, respectively. Using the sample slice without the lipid core, the maximum plaque stress values (unit: kPa) from the soft and median materials were 153.3 and 96.2, which are 67.7% and 5% higher than that (91.4) from the stiff material, while the maximum plaque strain values from the soft and median materials were 0.71 and 0.293, which are about 700% and 230% higher than that (0.089) from the stiff material, respectively. Without circumferential shrinkages, the maximum plaque stress values (unit: kPa) from the soft, median, and stiff models were inflated to 330.7, 159.2, and 103.6, which were 116%, 65%, and 13% higher than those from models with proper shrinkage. The effective Young's modulus from the 12 human carotid arteries studied varied from 137 kPa to 1435 kPa. The vessel circumferential shrinkage to the zero-pressure condition varied from 6% to 32%. The inclusion of proper shrinkage in models based on in vivo geometry is necessary to avoid over-estimating the stresses and strains by up 100%. Material stiffness had a greater impact on strain (up to 700%) than on stress (up to 70%) predictions. Accurate patient-specific material properties and circumferential shrinkage could considerably improve the accuracy of in vivo MRI-based computational stress/strain predictions. [DOI: 10.1115/1.4005685]

Keywords: carotid artery, atherosclerosis, MRI, Cine, plaque rupture

1 Introduction

It is widely accepted that atherosclerotic plaques with thin fibrous caps, large lipid cores, and inflammatory cell infiltrations are subject to local stress concentrations, which subsequently lead to fissured plaque, occlusive thrombi, and acute myocardial infarction and stroke [1–7]. Image-based computational models have been introduced to calculate critical plaque stress/strain conditions and assess plaque vulnerability [6–14]. However, the accuracy of computational stress and strain predictions is heavily dependent on the data and assumptions used by those models. The data needed in making plaque models can be roughly classified into three categories: (a) plaque morphology and components, (b) material properties, and (c) blood flow and pressure conditions. While recent developments in magnetic resonance imaging (MRI) techniques [1–3] have provided plaque morphology and flow data for in vivo MRI-based plaque models, noninvasive techniques to

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obtain patient-specific vessel material properties and shrinkage data are lacking in those models. This deficiency is a considerable limitation for in vivo image-based plaque models.

An extensive effort has been made by several research groups quantifying mechanical material properties of atherosclerotic arteries and their impact on stress/strain predictions. Holzapfel et al. investigated the layer- and direction-dependent ultimate tensile stress and stretch ratio of human atherosclerotic iliac arteries [15]. Anisotropic and highly nonlinear tissue properties were observed along with interspecimen differences. The adventitia demonstrated the highest strength and the fibrous cap in the circumferential direction showed low fracture stress. Different layers displayed different direction-dependent mechanical behaviors which are crucial for realistic computational models and accurate stress-strain predictions [8,15,16]. Williamson et al. investigated the sensitivity of wall stresses in diseased arteries to variable material properties using a histology-based model and reported that stresses in artery walls have low sensitivities for variation in the elastic modulus. Even a $\pm 50\%$ variation in elastic modulus leads to less than a 10% change in stress at the site of rupture [17]. Schulze-Bauer and Holzapfel proposed an approach of the determination of vessel constitutive equations from clinical data (i.e., limited pressure-diameter relations in the diastolic-systolic pressure range) by means of reasonable assumptions regarding in situ configurations and stress states of arterial walls. Human aorta samples were used to demonstrate the potential of their approach [18]. By using ex vivo MRI-based human carotid plaque models with fluid-structure interactions (FSI), Tang et al. investigated the impact of plaque structure and material properties on stress/strain behaviors and reported that softer materials led to higher stress values [19]. Lendon et al. showed that the weaker plaque caps appeared to be associated with increased macrophage activity [20,21]. Barrett et al. performed a radial indentation experiment on the human carotid artery, focusing on the quantification of the stiffness of the fibrous cap [22]. By summarizing previous experimental data along with their own results, they reported that the range of stiffness of the obtained fibrous cap tissues was very sparse, presumably due to the widely differing methods adopted and different types of arteries investigated. Teng et al. performed uniaxial tests on 73 axial and circumferential oriented adventitia, media, and intact specimens prepared from 6 human carotid arteries [23]. Their results showed that the mean axial and circumferential ultimate strength of the media group were 519 ± 270 kPa and 1230 ± 533 kPa, respectively, while the corresponding ultimate strength for the adventitia were 1996 ± 867 kPa and 1802 ± 703 kPa, respectively. Ohayon et al. investigated the influence of residual stress/strain on the biomechanical stability of vulnerable coronary plaques and their potential impact for evaluating the risk of plaque rupture [24]. By using 6 human coronary vulnerable plaque samples, opening angles were obtained and computational models were constructed based on histology. A wrapping-up technique was used to bring the opened vessel to its closed shape and then lumen pressure was applied to obtained the stress/strain predictions. Their results indicated that residual stress (as demonstrated by the opening angle) has considerable impact on stress/strain predictions. Most of the investigations were performed using ex vivo specimens and in vitro experimental techniques.

In addition to material properties and the previously discussed issues, another important issue for models based on in vivo data is the determination of a shrinkage rate (circumferential shrinkage for 2D models) needed to shrink the vessel in vivo geometry to its zero-pressure shape. In their in vivo MRI-based fluid-structure interaction (FSI) plaque models, Tang et al. and Yang et al. introduced a shrink-stretch process to: (a) shrink the vessel both axially and circumferentially to obtain the "no-load" shape as the numerical starting geometry, and (b) stretch and pressurize the vessel to recover its in vivo shape under pressure and stretch conditions [7,25–27]. The shrinkage in the axial direction was 9% so that the vessel would regain its in vivo length with a 10% axial stretch. The circumferential shrinkage for the lumen and outer wall was determined so that: (1) the total mass volume was conserved, and (2) the plaque geometry after 10% axial stretch and pressurization had the best match with the original in vivo geometry. No patient-specific data was available, therefore, the shrinkage and material properties were taken from the available literature. Speelman et al. and Gee et al. also demonstrated the necessity and importance of the preshrink process by using their in vivo computed tomography (CT)-based abdominal aortic aneurysm simulations [28,29]. Huang et al. proposed a nonuniform shrinking process which could achieve better agreement with in vivo vessel geometries [30]. By using 50 histological coronary sections (from 7 patients), perfusion fixed at 100 mm Hg, Speelman et al. demonstrated a backward incremental method to determine the zero pressure vessel geometry and initial stress conditions [31]. The neo-Hookean hyperelastic material properties from the literature were used in their study [32]. While the importance of the proper handling of prestress issues is well recognized, techniques to quantify the patient-specific shrinkage rate using in vivo data are missing in the current literature.

In this paper, noninvasive in vivo Cine and 3D multicontrast MRI data and modeling techniques were combined to obtain patient-specific carotid artery material properties and vessel circumferential shrinkage for zero-pressure geometry to improve model prediction accuracies. Our objectives are: (a) to introduce a noninvasive methodology based on in vivo MRI to quantify vessel material properties and the vessel shrinkage rate for zero-pressure geometry, (b) to noninvasively obtain in vivo a small patient-specific data set ($n = 12$) of human carotid vessel material properties and circumferential shrinkage rate, (c) to demonstrate the importance of including the circumferential shrinkage for models based on in vivo geometry, (d) to quantify the impact of material properties on the stress/strain predictions with and without circumferential shrinkage. These methods and results will fill a gap in the current literature. A 2D human carotid plaque model with and without a lipid core and three parameter sets representing stiff, median, and soft materials obtained from our patient data were used to demonstrate the impact of artery material stiffness on the computational stress and strain predictions.

2 Methods

2.1 MRI Data Acquisition. Twelve participants (10 male; 2 female; age 56–86; mean: 72) with carotid atherosclerosis were imaged by the Vascular Imaging Laboratory (VIL) of the University of Washington (UW) after patient consent, using protocols approved by the UW Institutional Review Board. Cuff systolic and diastolic arm pressure was recorded for modeling use. In vivo Cine and 3D multicontrast MR images of the carotid arteries were acquired using a 3.0 T whole-body scanner (Philips Achieva, R2.6.1, Best, The Netherlands) using a dedicated 8-channel, phased array carotid coil. The carotid bifurcation was located on 2D time of flight (TOF) and oblique black blood MR images. A 3.5 cm region centered on the carotid bifurcation was imaged by high-resolution axial bright and black blood imaging. Proton density (PD)-, T2- and T1-weighted images of carotid arteries were obtained. A 3D-TOF angiographic sequence and a magnetization prepared rapid gradient echo sequence for intraplaque hemorrhage detection were also performed. Imaging parameters were as follows: 3D TOF: TR/TE 20/4 ms, flip angle 20 deg; T1w: quadruple inversion-recovery, black-blood, 2D TSE, TR/TE 800/10 ms; T2w: multislice double IR (MDIR), TR/TE 4800/50 ms, PDw: multislice double IR (MDIR), TR/TE 4800/5 ms. All images were acquired using field of view 16 cm $\times$ 16 cm, acquisition matrix 256×256, slice thickness 2 mm, and in-plane resolution 0.6 mm $\times$ 0.6 mm. After machine interpolation, the in-plane resolution was 0.3 mm $\times$ 0.3 mm. Following the morphological evaluation, an electrocardiogram gated Cine sequence was used to measure vessel compliance. Eight slices (along the carotid bifurcation) were imaged with a temporal resolution of 30 cardiac phases. The MR images were segmented using a custom-made analysis tool

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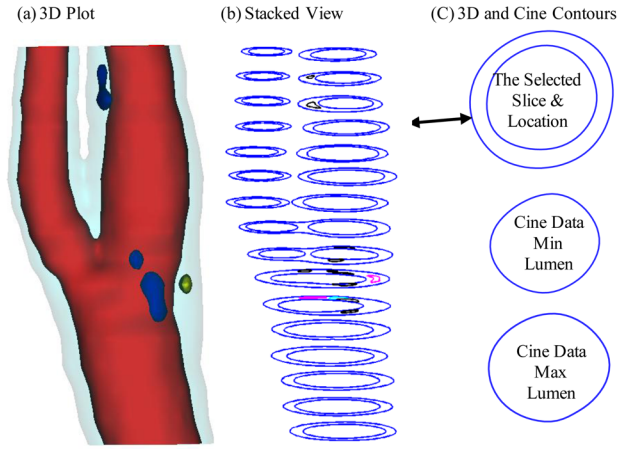


Fig. 1 (*a*) 3D geometry of a human carotid plaque sample reconstructed from MRI, (*b*) stacked contours, (*c*) selected slice from 3D MRI, Cine with maximum and minimum lumen, and Adina model output corresponding to maximum and minimum pressure

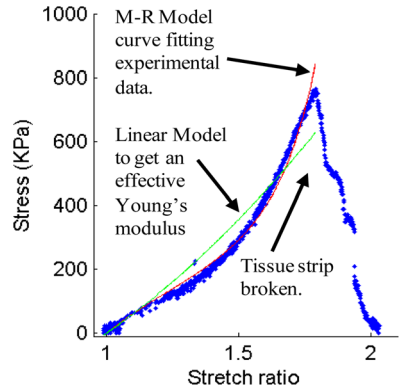


Fig. 2 Stress-stretch plot of uniaxial test of a circumferential strip sample from a human carotid artery. The red curve is from the Mooney-Rivlin model fitting experimental data. The green straight line is from the linear model fitting the experimental data. Parameter values of the Mooney-Rivlin model and linear model fitting the experimental data are: $c_1 = 94.6$ kPa, $D_1 = 6.81$ kPa, $c_2 = 0$, $D_2 = 2.0$, YM = 570 kPa.

(CASCADE developed at the VIL) to identify lipid-rich necrotic core, loose matrix, calcification, thrombus, hemorrhage, and ulceration when present [7]. For each patient, one location with the Cine sequence and the most nearly-circular slice from 3D MRI was selected for estimation of material parameter values in the modified Mooney-Rivlin model used in our previous publications. Figure 1 shows a sample plaque re-constructed from the MRI showing the 3D plaque geometry, stacked slices, and the location of the selected slice for the material parameter determination. "Plaque" is used to indicate an artery segment with considerable atherosclerosis. Figure 1 also provided the selected slice (Fig. 1(*c*)) and two plots of Cine lumen circumferences corresponding to minimum and maximum lumen pressure. The Cine data had only lumen contours and could not be used directly to make plaque models.

2.2 An Iterative Scheme Using Cine MRI to Quantify Vessel Material Parameter Values and Shrinkage Rate. The modified Mooney-Rivlin (M-R) material model was used for the vessel material whose strain energy function is given as follows [33,34]

$$W = c_1(I_1 - 3) + c_2(I_2 - 3) + D_1[\exp(D_2(I_1 - 3)) - 1] \quad (1)$$

$$I_1 = \Sigma C_{ii}, \quad I_2 = \frac{1}{2}\left[I_1^2 - C_{ij}C_{ij}\right] \quad (2)$$

where $\boldsymbol{C} = [C_{ij}] = \boldsymbol{X}^T\boldsymbol{X}$ is the right Cauchy-Green deformation tensor, I_1 and I_2 are the invariants of $\boldsymbol{C}$, $\boldsymbol{X} = [X_{ij}] = [\partial x_i/\partial a_j]$ is the deformation gradient, and c_1, c_2, D_1, and D_2 form the material parameter set M1 (or M2 as needed). The modified Mooney-Rivlin model was selected because it was able to fit carotid artery vessel properties measured by uniaxial mechanical testing data and good agreement was obtained (Fig. 2) [23]. The M-R model is able to cover the vessel material behaviors both near the no-load condition and the stiffening behavior under greater strain. For this paper, we have very limited data (only two data points): minimum and maximum pressure conditions with corresponding lumen circumference values. It is often the case for noninvasive in vivo studies that we have to work with limited data to derive patient-specific information for our models. Models with patient-specific material properties and shrinkage will be an improvement over models using those data from the literature. Choosing the M-R model helps us to obtain a better estimate of the zero-pressure lumen circumference and the stiffening behavior, compared to using a linear model. The selection of parameters is not unique. In this study, we set $c_2 = 0$ and $D_2 = 2$, according to our previous experiences [7,13]. We also set $c_1/D_1 = 368/144$ [7] to simplify the numerical iterative search procedure. In summary, the selection of models and parameters was based on our prior knowledge that the M-R model fits well to the extensive experimental data obtained from human carotid samples ex vivo. Those selections are by no means unique. Future improvements for better parameter determinations are possible when more data points become available.

An iterative procedure (see Fig. 3 for details) was followed to adjust the parameter values in the modified Mooney-Rivlin model and the circumferential shrinkage rate to match both te maximum and minimum Cine lumen circumferences corresponding to patient-specific systolic and diastolic pressures, respectively. The procedure starts from initial estimates of parameter values c_1 (36.6 kPa) and D_1 (14.4kPa) and the lumen shrinkage rate S1 (10%) [7]. The original in vivo lumen contour was shrunk by S1 using its centroids as its local coordinate origin. The outer-boundary shrinkage rate was determined using conservation of the vessel area. Since the slice was near-circular, the lumen contour and outer-boundary of the vessel could be approximated by circles whose radii could be easily calculated from their circumferences. To keep the vessel wall area unchanged (the conservation law of mass is achieved by

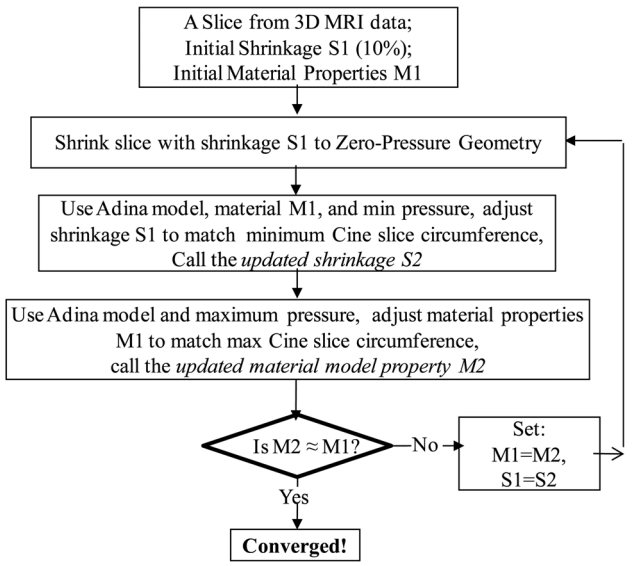


Fig. 3 The iterative procedure to determine material parameters and vessel shrinkage rate

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conservation of the area for 2D models), the outer-boundary shrinkage S_{out} was calculated by solving this equation

$$\pi\left[(R_{out}(1-S_{out}))^2-(R_{lumen}\times(1-S_{lumen}))^2\right]=\pi\left[(R_{out})^2-(R_{lumen})^2\right]=\text{Const} \quad (3)$$

where R_{out} and R_{lumen} denote the in vivo radii of the outer boundary and lumen before preshrinkage, respectively. Here, S_{lumen} was denoted by S1 or S2 in the iterative procedure. The outer-boundary was shrunk using S_{out} determined from Eq. (3). The Area conservation was double-checked by numerical area calculations after the shrinkage and vessel area was very well conserved.

At each iteration step, after we obtained the zero-pressure slice geometry, a finite element model was constructed and solved by a commercial software ADINA (ADINA R & D, Watertown, MA). A constant pressure boundary condition was imposed on the lumen using either systolic or diastolic arm pressures. The external pressure was set to zero. A typical 2D model has about 500–1000 9-node 2D-solid elements, depending on the complexity of the geometry and components. The lumen circumference from the pressured slice was compared with the in vivo shape to see if the desired accuracy is achieved. The iterative procedure was set to stop if the relative changes of the shrinkage and parameter values were less than 0.01 (1%). Each iteration included adjustment of the shrinkage rate and material parameter values, adjustment of the geometry, the finite element model construction and re-mesh, and the solution of the model. This took about 1–2 h. Each patient case took about 20–50 iterations to reach the desired tolerance. The analysis for one patient case needed a few days to finish. The CPU time for the solution of each 2D plaque model is only 1–2 min using a 16-processor parallel computer.

2.3 Using 2D Structure-Only Models to Quantify the Effect of Material Properties on Stress/Strain Distributions. Using a sample 2D plaque slice whose geometry will be presented in the Results section, 2D structure-only models with and without a lipid core were used to demonstrate the impacts of the variation of the material parameters on the plaque stress and strain distributions which, are the maximum principal stress and strain denoted by Stress-P_1 and Strain-P_1, respectively. We also denote these as "stress" and "strain" when the meaning is clear. As a first approximation, the vessel material and the lipid core were assumed to be hyper-elastic, isotropic, incompressible, and homogeneous. The modified Mooney-Rivlin model was used for all materials. The lumen pressure was set as 120 mm Hg. Three material parameter sets obtained from our patient data representing the stiffest, median, and softest materials were used for the vessel: (a) stiff material obtained from Patient 3 (P3), $c_1=90$ kPa, $D_1=35.2$ kPa, $D_2=2$, (b) median material from Patient 4 (P4), $c_1=27.0$ kPa, $D_1=10.57$ kPa, $D_2=2$, and (c) soft material from Patient 11 (P11), $c_1=8.06$ kPa, $D_1=3.37$ kPa, $D_2=2$. Here, c_2 was set to zero for all cases. Parameter values for the lipid core in our previous publications were used: $c_1=0.5$ kPa, $D_1=0.5$ kPa, $D_2=0.5$ [7,35]. The 2D plaque models were solved by a commercial finite element package ADINA (ADINA R & D, Watertown, MA) and the results are presented in Sec. 3. Simulations were also run with and without circumferential shrinkage to demonstrate the significance of the preshrink process for in vivo image-based models.

2.4 Obtaining an Effective Young's Modulus as a Stiffness Indicator for the Material. For comparison purposes, it is desirable to have a single parameter to compare vessel stiffness from different patients. The stress-stretch relationship for the Mooney-Rivlin model is given by

$$\sigma=\sigma_3=\lambda\frac{dW}{d\lambda}=2\lambda(\lambda-\lambda^{-2})\left(c_1+D_1D_2e^{D_2\left(\lambda^2+2/\lambda-3\right)}\right) \quad (4)$$

where σ is the Cauchy stress, and λ is the stretch ratio. The effective Young's modulus (YM) E for the stretch ratio interval [1.0,1.3] is defined as (note: σ is the Cauchy stress)

$$\sigma=E(\lambda-1)\lambda \quad (5)$$

A least-squares technique was used to determine the E values so that the linear model (5) has the best fit to the Mooney-Rivlin model for each plaque.

3 Results

3.1 Patient-Specific Material Properties. Material model parameters for the 12 participants were obtained using the procedure given in Sec. 2.2. The Mooney-Rivlin model parameter values, maximum and minimum Cine lumen circumferences, maximum and minimum lumen pressure, effective YM, and circumferential shrinkage for the 12 plaque samples are given in Table 1. Strong correlations were observed: (a) positive correlation between Max-P and YM ($r=0.795$, $p<0.004$), (b) negative correlation between Max-P and shrinkage rate ($r=-0.403$, $p<0.0001$, and (c) negative correlation between YM and shrinkage ($r=-0.673$, $p<0.002$). Stress-stretch plots based on Eq. (3) for all of the cases are presented in Fig. 4. It can be seen that the YM for the stiffest case (P3) was nearly 10 times higher than that for the softest case (P11). The averaged YM value from the 12 cases was 479 kPa, which was consistent with the current literature [6,7,15,23,26]. It should be noted that the three patients with systolic pressure greater than 140 mm Hg had noticeably stiffer vessels and a smaller shrinkage

Table 1 Material parameter values in modified Mooney-Rivlin models and shrinkage for carotid atherosclerotic plaques from 12 patients based on Cine MRI data. $D_2=2$ for all cases.

Patients	Max-Cir (cm)	Min-Cir (cm)	Max-P (mm Hg)	Min-P (mm Hg)	C_1 (kPa)	D_1 (kPa)	YM (kPa)	Circumference shrinkage
P1	2.30	1.99	133	80	18.00	7.05	287.4	0.185
P2	2.24	1.95	100	60	12.90	5.05	205.8	0.167
P3	3.37	3.12	175	75	90.00	35.22	1435.9	0.060
P4	1.76	1.60	130	70	27.00	10.57	430.8	0.126
P5	1.51	1.38	110	60	18.75	7.34	299.1	0.111
P6	1.91	1.73	120	70	16.90	6.62	269.8	0.145
P7	1.84	1.74	120	68	40.20	15.77	642.3	0.062
P8	2.69	2.42	125	54	35.00	13.70	558.4	0.070
P9	2.84	2.53	160	70	38.00	14.94	607.9	0.080
P10	1.88	1.70	146	66	41.50	16.27	662.8	0.078
P11	2.04	1.74	120	70	8.60	3.37	137.2	0.326
P12	1.32	1.22	84	50	13.00	5.10	207.6	0.10
Average	2.14	1.93	127	66	29.99	11.75	478.8	0.118

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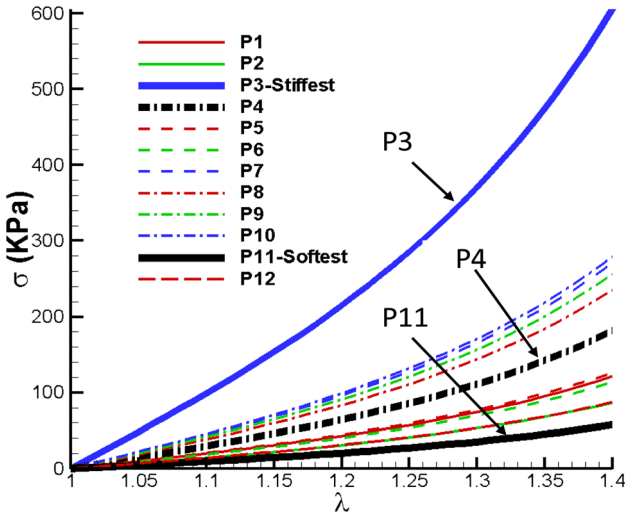


Fig. 4 Stress-stretch curves from Mooney-Rivlin models using parameter values determined from Cine MRI for the 12 patients studied

rate. The patient for the stiffest case had a large span between his diastolic and systolic pressures (75–175 mm Hg).

3.2 Impact of Material Parameters on Stress/Strain Predictions Using 2D Models With Proper Circumferential Shrinkage. Figures 5 and 6 show the stress and strain plots of the sample 2D plaque slice using three different parameter sets corresponding to: (a) stiff material from Patient 3, effective YM = 1436 kPa, (b) median material from Patient 4, effective YM = 430.8 kPa, and (c) soft material from Patient 11, effective YM = 137.2 kPa. Roughly, the YM value of the stiff material was about three times that from the median material, and the YM value of the median material was about three times that from the soft material. A preshrink process was applied to each case to obtain the zero-pressure vessel geometry so that the pressurized vessel geometry matched the original in vivo geometry as closely as possible. The maximum stress and strain values for all cases and preshrink percentages are summarized in Table 2.

When the lipid core was included, the maximum stress/strain values were found at the thin cap covering the lipid core (Figs. 5 and 6). The maximum stress value from the soft case (Fig. 5(*d*)) was 216.7 kPa, which was 33% higher than in the stiff case (Fig. 5(*b*)). *The increase of the maximum stress from the stiff case (P3) to the median case (P4) was only 4%. Noticing that the effective YM value for P3 was about 230% higher than that for P4, this came as quite a surprise*. For the cases without the lipid core, the maximum stress from the soft case (P11) was 153.3 kPa, which is 68% higher than that from the stiff case. The increase of the maximum stress from the stiff case to the median case was, again, only 5%. The effect of the material stiffness on the plaque stress was clearly nonlinear, with more noticeable changes observed for softer materials.

The impact of material stiffness on strain was much greater. With the lipid core, the maximum strain from the soft case (Fig. 6(*d*)) was 0.793, which was 454% higher than that (strain = 0.143) from the stiff case (Fig. 6(*b*)). The increase of the maximum strain from the stiff case to the median case was 194%, which is much greater than the percentage for the corresponding stress comparison. For the cases without the lipid core, the maximum strain from the soft case was 0.711, which is about 700% higher than in the stiff case (0.089). The increase of the maximum strain from the stiff case to the median case was 229%. Shrinkages were 5–6% (P3), 13–15% (P4), and 33–34.5% (P11), respectively (Table 2).

3.3 Models Without Shrinkage Gave Over-Estimated Plaque, Wall Stress, and Strain Predictions. The preshrink process to obtain zero-pressure plaque geometry is well justified and easy to accept. However, the errors in the stress/strain predictions caused by models without proper shrinkage were normally demonstrated in previous publications using hypothetical shrinkage (normally 8–10%) and material properties (just one set of parameters for all patients in one paper) from the literature [25–27,30]. Shrinkage is linked to material properties. Different material stiffness needs different shrinkage so that in vivo geometry could be recovered under pressure. With our new artery shrinkage and material data, we can compare results from models with and without shrinkage to quantify these errors caused by models that do not utilize the shrinking process. By using three materials, we are also demonstrating the need for different shrinkage rates. Figures 7 and 8 give plaque stress and strain plots from models without shrinkage, parallel to the plots in Figs. 4 and 5 obtained from models with proper

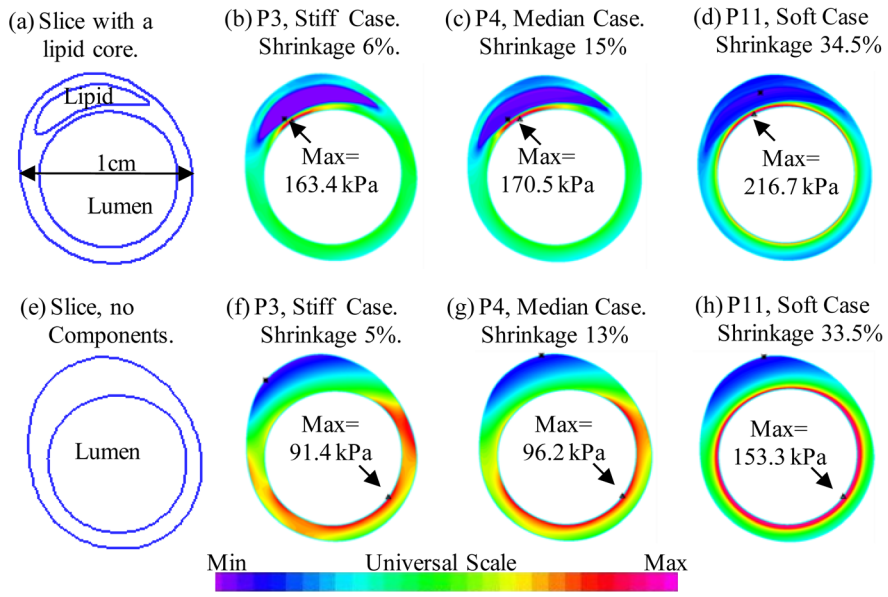


Fig. 5 Plots of plaque stress distributions showing the effect of material stiffness on stress predictions

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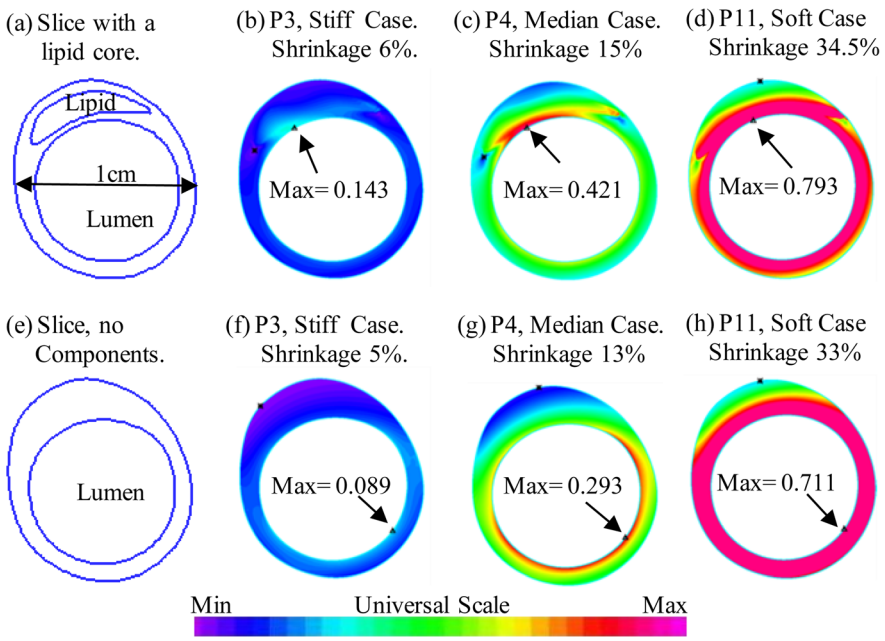


Fig. 6 Plots of plaque strain distributions showing that the material stiffness has much greater impact on strain predictions. Predicted maximum strain values for softer materials were 200–700% higher than those from the stiff material. Uniform color scales were applied for (*b*)–(*d*) and (*f*)–(*h*), respectively.

shrinkages. A comparison of the maximum stress/strain values from the six cases are given in Table 2. Without circumferential shrinkages, the maximum plaque stress values (unit: kPa) from the soft, median, and stiff models without a lipid core were inflated to 330.7, 159.2, and 103.6, which were 116%, 65%, and 13% higher than those from models with proper shrinkage. The predicted maximum strain values from the soft, median, and stiff models without a lipid core and shrinkage were 0.901, 0.405, and 0.098, compared to 0.711, 0.293, and 0.089 from models with proper shrinkage.

Using models without shrinkage, a falsely greater impact of material stiffness on plaque stress/strain predictions would be reported. For the cases including the lipid core, the maximum stress value from the soft case (Fig. 7(*d*)) was 353.6 kPa, which was 100% higher than that from the stiff case (Fig. 7(*b*)), compared to a mere 33% increase in Fig. 5. The increase of the maximum stress from the stiff case to the median case was 32%, compared to a 4% increase in Fig. 5. The differences in the plaque strain predictions were smaller. The maximum strain value increases were 500% (from the stiff model to the soft model) and 224% (from the stiff model to the median model), compared to 454% and 194% from Fig. 6. These result show that neglecting shrinkage would result in predicting a much greater impact of material stiffness on plaque stress/strain values.

4 Discussion

4.1 Vessel Material Stiffness Has Greater Impact on Strain Predictions. Using the no-lipid models, the maximum strain values from the soft and median models were 700% and 230% higher than that from the stiff material, while the maximum stress values were only 70% and 5% higher. Considering that most research reports have been focused on critical stress conditions, our results indicated that plaque mechanical investigations should include both critical stress and strain conditions when accurate in vivo vessel material properties become available.

4.2 Impact of Vessel Material Stiffness on Stress Was Nonlinear and Was Very Modest Near the Median Stiffness Range. The effective YM value from the stiff case was 1436 kPa, which is 230% higher than that for the median case. However, the predicted maximum stress values differed by a mere 4–5%. Then from the median case to the soft case, the difference went up to 70%.

Table 2 Summary of maximum stress and strain values for all cases showing impact of material stiffness and pre-shrink process on stress/strain predictions. Pre-shrink percentages were included for references.

		Maximum stress values (kPa)			Maximum strain values		
Shrink/No shrink	Lipid/No lipid	P3 Stiff	P4 Median	P11 Soft	P3 Stiff	P4 Median	P11 Soft
With shrinkage	With lipid	163.4	170.5	216.7	0.143	0.421	0.793
		100%	*104%*	*133%*	*100%*	*294%*	*554%*
	Pre-shrink	*6%*	*15%*	*34.5%*	*6%*	*15%*	*34.5%*
	No lipid	91.4	96.2	153.3	0.089	0.293	0.711
		100%	*105%*	*168%*	*100%*	*329%*	*798%*
	Pre-shrink	*5%*	*13%*	*33%*	*5%*	*13%*	*33%*
No shrinkage	With lipid	176.4	232.9	353.6	0.153	0.496	0.917
		100%	*132%*	*200%*	*100%*	*324%*	*599%*
	No lipid	103.6	159.2	330.7	0.098	0.405	0.901
		100%	*154%*	*319%*	*100%*	*413%*	*919%*

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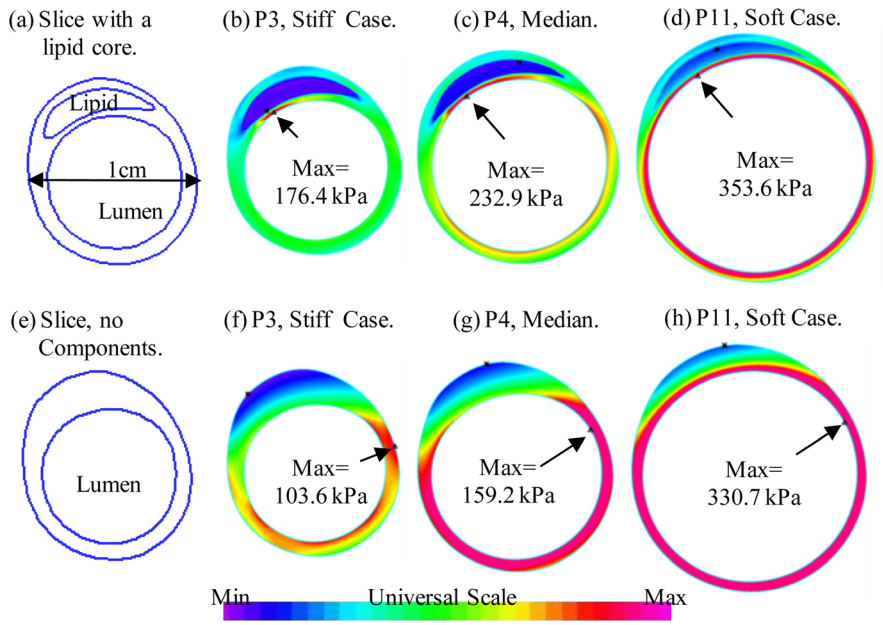


Fig. 7 Plots of plaque stress (Stress-P_1) from models without preshrinkage gave inflated plaque stress predictions. Uniform color scales were applied for (*b*)–(*d*) and (*f*)–(*h*), respectively.

4.3 Preshrink Process Has Important Impact on Stress/Strain Predictions. It is commonly accepted that material properties have considerable impact on stress/strain predictions. Surprisingly, when proper shrinkage was applied, since it was required to match the in vivo vessel morphology under pressure, maximum plaque stress values between the stiff and median cases differed only by 4–5%, where the YM value of the stiff material was about three times that of the median material. Without proper shrinkage, maximum stress from the median case without lipid was 54% higher than that from the stiff case. Using the models without lipid, with proper shrinkage, the maximum stress from the soft case was 68% higher than that from the stiff material, compared to a *220% increase* for the same case without shrinkage. Similar comparisons can be found from Table 2 for the strain predictions. This gives us a strong indication that computational models should include proper shrinkage to provide accurate stress/strain predictions.

4.4 Importance of Residual Stress, Circumferential, and Axial Shrinkage, and Balancing Effort and Model Assumptions. As demonstrated by Ohayon et al., the residual stress has a huge impact on the fibrous cap stress. Ohayon et al. showed that the predicted peak cap stress was about 50 kPa with the residual-stress model but could reach 220 kPa using the nonresidual-stress model [24]. Such a huge difference certainly warrants the necessity of considering residual-stress in plaque stress analysis. However, an artery segment under in vivo

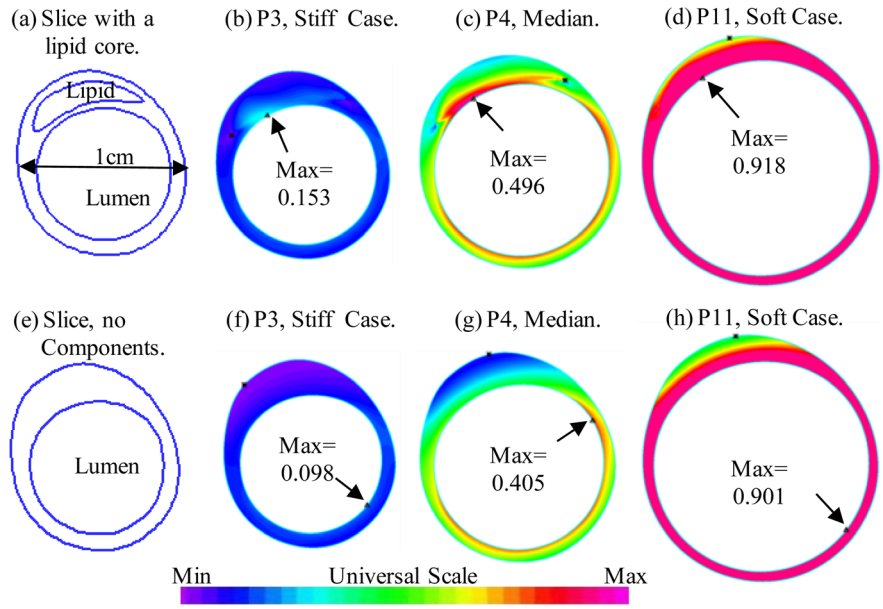


Fig. 8 Plots of plaque strain (Strain-P_1) from models without preshrinkage gave inflated plaque strain predictions. Uniform color scales were applied for (*b*)–(*d*) and (*f*)–(*h*), respectively.

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condition goes through three stages to its open-up shape ex vivo: axial shrinkage (up to 30–50%), circumferential shrinkage (5–20% based on our data), and final opening-up with an opening angle. Our initial opening angle results from human coronary arteries (mean α; =120 deg, n = 4) and carotid arteries (mean α; =63.5 deg, n = 5) were reported in Table 2 of Ref. [37]. The axial shrinkage and circumferential shrinkage, which are of equal importance compared to the opening angle step but far less well known, were brought up in our earlier papers [25–27]. The opened shape is assumed to be stress-free, which is a well-accepted approximation. Omitting any of the stages would lead to stress/strain prediction errors on the order of 50%–100% or even more. Our effort in quantifying patient-specific circumferential shrinkage is similar to the effort quantifying the open angle addressing the residual stress issue.

Considerable simplifications and assumptions have to be made for models based on in vivo data which is scarce in the current literature, especially for human data. Most current in vivo plaque models only have morphology from medical images. Patient-specific material properties, on-site pressure, axial shrinkage, circumferential shrinkage, and opening angle (for 3D, we need an opening angle for many locations) are not currently available. The method introduced in this paper addresses the vessel material properties and circumferential shrinkage issues, subject to its limitations and assumptions. Results from computational models should always be interpreted with caution and with their model assumptions kept in mind. With reasonable validations, computational stress/strain predictions are still of value in plaque assessment since all patients would be subjected to the same assumptions. Better models would be employed as technology and methods evolve.

4.5 2D Versus 3D Approach. Cine data were acquired at multiple locations for each patient. Our acquired data indicated that vessel stiffness varied from slice to slice for each plaque, affected by location, plaque composition, and stenosis severity. To use a 3D approach, 3D plaque models would need to be constructed. The iterative process would require that the 3D model geometry be modified between 20–50 times or more (it is harder to obtain a 3D match) to reach the proper shrinkage and material parameters for all slice locations. The 3D model would also need to have different material parameter values for each slice. Nonuniform material models would be needed to accommodate the slice-to-slice variations. That parameter determination process would require very considerable effort for each plaque, which was not practical to adopt. Using our 2D approach, it took several days to adjust the 2D model 20–50 times to achieve convergence for both circumferential shrinkage rate and parameter values. The 2D approach represents a reasonable first-step approximation for the material parameter quantification. With further automation, it has the potential for direct clinical application and software commercialization.

4.6 Gross Vessel Material Properties and Plaque Component Material Properties. It is clearly of great important to be able to noninvasively quantify patient-specific plaque component properties under in vivo conditions. However, it should be noted that the Cine data only provided lumen contours which gave us data to have a first-order estimation of gross vessel material properties, which fills a gap in the current modeling approach. The Cine and 3D multicontract MRI fit naturally together and are clinically viable. However, we are working with very limited data. The Cine MRI does not have plaque component information to derive component material properties. In vitro experimental techniques or other high-resolution modalities could be employed to quantify plaque component properties.

4.7 Impact of Hypertension. Hypertension may be considered a common thread in most cardiovascular diseases. Table 1 shows that the three patients who had higher than 140 mm Hg maximum pressure all had higher vessel stiffness and smaller shrinkage rate. Our focus in this paper is on vessel material properties and shrinkage. Pressure condition was used to determine vessel material properties and shrinkage, which in turn, would influence stress/strain calculations. Our earlier results indicated that a 50% pressure increase could lead to a 100% increase in critical stress values [25]. Since we do not have direct pressure measurements at the present time, we plan to investigate the impact of hypertension when better pressure data becomes available.

4.8 Validations. The procedure of using Cine MRI to determine vessel material properties is self-validated in the sense that the material parameter values were used to match in vivo plaque geometries under both systolic and diastolic pressure conditions. Cine MRI is widely accepted to acquire time-dependent vessel motion and deformations, with the understanding that the MRI resolution needs improvement. The ultimate validation of the procedure would be to obtain the tissue samples from the patients and perform in vitro mechanical testing. However, this was not possible since no surgical procedures were performed on those patients.

Uniaxial mechanical testing of human carotid circumferential and axial strips was performed in our previous studies using 73 strips from 6 artery samples with AHA Type III lesions [23]. Figure 2 presents the stress-stretch plot of the uniaxial test results of one circumferential strip sample from a human carotid artery showing that the modified Mooney-Rivlin models (red lines in the plot) provided a good fit to the experimental data and the effective YM values were consistent with our findings in this paper. The effective YM values would be lower if the stretch interval [1.0,1.3] was used as in Table 1 ([1.0,1.3] was more relevant to the Cine data there). Again, the linear model was used only to get a single parameter value as an easy indicator of vessel stiffness. Clearly, the Mooney-Rivlin models provided a better fit to the experimental data, compared to the linear models.

4.9 Other Limitations. The multilayer structure and anisotropic properties of arteries were not considered since Cine data provided only circumference variations under cardiac pressure. Another limitation was that arm cuff systolic/diastolic blood pressures were used as the on-site pressure because location-specific pressure measurement was not available. Currently, arm cuff pressure values were used in most image-based publications, with the belief that they are suitable approximations to on-site actual pressure. Since only two pressure values were used, a phasic difference at the two locations (arm and carotid) would not affect our results. If we could have on-site pressure profiles, we could match pressure values with Cine data which has 30 time points. These 30 matched (lumen circumference, pressure) data points would give us a better data set to determine vessel material properties.

5 Conclusion

A novel noninvasive approach of combining Cine MRI, multicontract 3D MRI, and computational modeling was introduced to quantify patient-specific carotid artery material properties and vessel circumferential shrinkage under in vivo conditions. This technique fills a gap in the current literature. The average effective Young's modulus (YM) for the 12 patients studied was 478 kPa. Using a 2D plaque model (with and without a lipid core) and representative stiff (effective YM = 1436 kPa), median (effective YM = 431 kPa), and soft (effective YM = 137 kPa) materials obtained from three patients, the maximum plaque wall stress value from the soft material was only 30–70% higher than that from the stiff material, while the maximum plaque wall strain value from the soft material was about 450%–700% higher than that from the stiff material, with or without a lipid core. Circumferential shrinkages for the soft, median, and stiff cases (without a lipid core) were 33%, 13%, and 5%, respectively. Computational

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plaque models should include proper shrinkage to obtain accurate stress/strain predictions. The approach combining Cine MRI with modeling and multicontrast MRI provides an opportunity to include patient-specific vessel material properties and circumferential shrinkage in computational models, which could considerably improve the accuracy of model stress/strain predictions.

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References

[1] Underhill, H. R., Hatsukami, T. S., Fayad, Z. A., Fuster, V., and Yuan, C., 2010, "MRI of Carotid Atherosclerosis: Clinical Implications and Future Directions," Nat. Rev. Cardiol., **7**(3), pp.165–73.

[2] Yuan, C., Zhang, S. X., Polissar, N. L., Echelard, D., Ortiz, G., Davis, J. W., Ellington, E., Ferguson, M. S., and Hatsukami, T. S., 2002, "Identification of Fibrous Cap Rupture with MRI Is Highly Associated with Recent Transient Ischemic Attack or Stroke," Circulation, **105**, pp.181–185.

[3] Yuan, C., Mitsumori, L. M., Beach, K. W., Maravilla, K. R., 2001, "Special Review: Carotid Atherosclerotic Plaque: Noninvasive MR Characterization and Identification of Vulnerable Lesions," Radiology, **221**, pp. 285–299.

[4] Fuster, V., Cornhill, J. F., Dinsmore, R. E., Fallon, J. T., Insull, W., Libby, P., Nissen, S., Rosenfeld, M. E., and Wagner, W. D., eds., 1998, *The Vulnerable Atherosclerotic Plaque: Understanding, Identification, and Modification*, AHA Monograph Series, Futura Publishing, Armonk, NY.

[5] Casscells, W., Naghavi, M., and Willerson, J.T., 2003, "Vulnerable Atherosclerotic Plaque: A Multifocal Disease," Circulation **107**, pp. 2072–2075.

[6] Friedman, M. H., Krams, R., and Chandran, K. B., 2010, "Flow Interactions with Cells and Tissues: Cardiovascular Flows and Fluid-Structure Interactions," Ann. Biomed. Eng., **38**(3), pp. 1178–1187.

[7] Tang, D., Teng, Z., Canton, G., Yang, C., Ferguson, M., Huang, X., Zheng, J., Woodard, P. K., and Yuan, C., 2009, "Sites of Rupture in Human Atherosclerotic Carotid Plaques Are Associated with High Structural Stresses. An In Vivo MRI-Based 3D Fluid-Structure Interaction Study," Stroke, **40**, pp. 3258–3263.

[8] Holzapfel, G. A., Stadler, M., and Schulze-Bause, C. A. J., "A Layer-Specific Three-Dimensional Model for the Simulation of Balloon Angioplasty Using Magnetic Resonance Imaging and Mechanical Testing," Ann. Biomed. Eng., **30**(6), pp. 753–767.

[9] Bluestein, D., Alemu. Y., Avrahami, I., Gharib, M., Dumont, K., Ricotta, J. J., and Einav, S., 2008, "Influence of Microcalcifications on Vulnerable Plaque Mechanics Using FSI Modeling," J. Biomech., **41**(5), pp. 1111–1118.

[10] Teng, Z., Canton, G., Yuan, C., Ferguson, M., Yang, C., Huang, X., Zheng, J., Woodard, P. K., Tang, D., 2010, "3D Critical Plaque Wall Stress is a Better Predictor of Carotid Plaque Rupture Sites than Flow Shear Stress: An In Vivo MRI-Based 3D FSI Study," J. Biomech. Eng, **132**(3), p. 031007.

[11] Kaazempur-Mofrad, M. R., Isasi, A. G., Younis, H. F., Chan, R. C., Hinton, D. P., Sukhova, G., Lamuraglia, G. M., Lee, R. T., and Kamm, R. D., 2004, "Characterization of the Atherosclerotic Carotid Bifurcation Using MRI, Finite Element Modeling, and Histology," Ann. Biomed. Eng., **32**(7), pp. 932–946.

[12] Suo, J., Oshinski, J. N., and Giddens, D. P., 2008, "Blood Flow Patterns in the Proximal Human Coronary Arteries: Relationship to Atherosclerotic Plaque Occurrence," Mol Cell Biomech., **5**(1):9–18.

[13] Tang, D., Yang, C., Zheng, J., Woodard, O. K., Sicard, G. A., Saffitz, J. E., and Yuan, C., 2004, "3D MRI-Based Multi-Component FSI Models for Atherosclerotic Plaques a 3-D FSI model," Ann. Biomed. Eng., **32**(7), pp. 947–960.

[14] Li, Z. Y., Howarth, S., Trivedi, R. A., U-King-Im, J. M., Graves, M. J., Brown, A., Wang, L. Q., and Gillard, J. H., 2006, "Stress Analysis of Carotid Plaque Rupture Based on In Vivo High Resolution MRI," J. Biomech., **39**, pp. 2611–2622.

[15] Holzapfel, G. A., Sommer, G., and Regitnig, P., 2004, "Anisotropic Mechanical Properties of Tissue Components in Human Atherosclerotic Plaques," ASME J. Biomech. Eng., **126**(5), pp. 657–665.

[16] Holzapfel, G. A., Gasser, T. C., and Ogden, R. W., 2000, "A New Constitutive Framework for Arterial Wall Mechanics and a Comparative Study of Material Models," J. Elast., **61**, pp. 1–48.

[17] Williamson, S. D., Lam, Y., Younis, H. F., Huang, H., Patel, S., Kaazempur-Mofrad, M. R., and Kamm, R. D., 2003, "On the Sensitivity of Wall Stresses in Diseased Arteries to Variable Material Properties," ASME J. Biomech. Eng., **125**, pp. 147–155.

[18] Schulze-Bauer, C. A. J. and Holzapfel, G. A., 2003, "Determination of Constitutive Equations for Human Arteries from Clinical Data," J. Biomech., **36**, pp. 165–169.

[19] Tang, D., Yang, C., Zheng, J., Woodard, P. K., Saffitz, J. E., Sicard, G. A., Pilgram, T. K., and Yuan, C., 2005, "Quantifying Effects of Plaque Structure and Material Properties on Stress Behaviors in Human Atherosclerotic Plaques Using 3D FSI Models," ASME J. Biomech. Eng., **127**(7), pp. 1185–1194.

[20] Lendon, C. L., Davis, M. J., Born, G. V. R., and Richardson, P.D., 1991, "Atherosclerotic Plaque Caps are Locally Weakened when Macrophage Density Is Increased," Atherosclerosis, **87**, pp. 87–90.

[21] Lendon, C. L., Davies, M. J., Richardson, P. D., and Born, G. V., 1993. "Testing of Small Connective Tissue Specimens for the Determination of the Mechanical Behavior of Atherosclerotic Plaques," J. Biomech. Eng., **15**(1), 27–33.

[22] Barrett, S. R. H., Sutcliffe, M. P. F., Howarth, S., Li, Z-Y., and Gillard, J. H., 2009, "Experimental Measurement of the Mechanical Properties of Carotid Atherothrombotic Plaque Fibrous Cap," J. Biomech., **42**(11), pp. 1650–1655.

[23] Teng, Z., Tang, D., Zheng, J., Woodard, P. K., and Hoffman, A. H., 2009, "An Experimental Study on the Ultimate Strength of the Adventitia and Media of Human Atherosclerotic Carotid Arteries in Circumferential and Axial Directions," J. Biomech., **42**(15), pp. 2535–2539.

[24] Ohayon, J., Dubreuil, O., Tracqui, P., Floc'h, S. L., Rioufol, G., Chalabreysse, L., Thivolet, F., Pettigrew, R. I., and Finet, G., 2007, "Influence of Residual Stress/Strain on the Biomechanical Stability of Vulnerable Coronary Plaques: Potential Impact for Evaluating the Risk of Plaque Rupture," Am. J. Physiol. Heart Circ. Physiol., **293**, pp. H1987–H1996.

[25] Tang, D., 2006, "Modeling Flow in Healthy and Stenosed Arteries," *Wiley Encyclopedia of Biomedical Engineering*, M. Akay, ed., John Wiley & Sons, Inc., Hoboken, NJ, pp. 1–16, 2006.

[26] Tang, D., Yang, C., Mondal, S., Liu, F., Canton, G., Hatsukami, T. S., and Yuan, C., 2008, "A Negative Correlation Between Human Carotid Atherosclerotic Plaque Progression and Plaque Wall Stress: In Vivo MRI-Based 2D/3D FSI Models," J. Biomech., **41**(4), pp. 727–736.

[27] Yang, C., Tang, D., Yuan, C., Hatsukami, T. S., Zheng, J., and Woodard, P. K., 2007, "In Vivo/Ex Vivo MRI-Based 3D Models with Fluid-Structure Interactions for Human Atherosclerotic Plaques Compared with Fluid/Wall-Only Models," CMES: Computer Modeling in Engineering and Sciences, **19**(3): 233–245.

[28] Speelman, L., Bosboom, E. M., Schurink, G. W., Buth, J., Breeuwer, M., Jacobs, M. J., and van de Vosse, F. N., 2009, "Initial Stress and Nonlinear Material Behavior in Patient-Specific AAA Wall Stress Analysis," J. Biomech., **42**(11), pp. 1713–1719.

[29] Gee, M. W., Reeps, C., Eckstein, H. H., and Wall, W. A., 2009, "Prestressing in Finite Deformation Abdominal Aortic Aneurysm Simulation," J. Biomech., **42**(11), pp. 1732–1739.

[30] Huang, Y., Teng, Z., Sadat, U., Hilborne, S., Young, V. E., Graves, M. J., Gillard, J. H., 2011, "Non-Uniform Shrinkage for Obtaining Computational Start Shape for In-Vivo MRI Based Plaque Vulnerability Assessment," J. Biomech., **44**(12), pp. 2316–2319.

[31] Speelman, L., Akyildiz, A. C., den Adel, B., Wentzel, J. J., van der Steen, A. F., Virmani, R., van der Weerd, L., Jukema, J. W., Poelmann, R. E., van Brummelen, E. H., and Gijsen, F. J., 2011, "Initial Stress in Biomechanical Models of Atherosclerotic Plaques," J. Biomech., **44**(13), pp. 2376–2382.

[32] Holzapfel, G. A., 2000, *Nonlinear Solid Mechanics: A Continuum Approach for Engineering*, Wiley, Chichester, New York, 2000.

[33] Bathe, K. J., 1996, *Finite Element Procedures*, Prentice Hall, Inc., New Jersey.

[34] Bathe, K. J., ed., 2002, *Theory and Modeling Guide, Vols. I and II: ADINA and ADINA-F*, ADINA R & D, Inc., Watertown, MA.

[35] Yang, C., Bach, R., Zheng, J., El Naqa, I., Woodard, P. K., Teng, Z. Z., Billiar, K. L., and Tang, D., 2009, "In Vivo IVUS-Based 3D Fluid Structure Interaction Models With Cyclic Bending and Anisotropic Vessel Properties for Human Atherosclerotic Coronary Plaque Mechanical Analysis," IEEE Trans. Biomed. Eng., **56**(10), pp. 2420–2428.

[36] Humphrey, J. D., 2002, *Cardiovascular Solid Mechanics*, Springer-Verlag, New York.

[37] Kural, M. H., Cai, M., Tang, D., Gwyther, T., Zheng, J., and Billiar, K. L., 2011, "Planar Biaxial Characterization of Diseased Human Coronary and Carotid Arteries for Computational Modeling," J. Biomech. (in press).

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